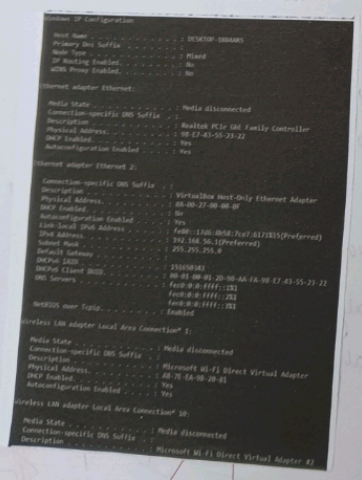




Expt no: 01
Date: 16/07/26

BASIC NETWORKING COMMANDS IN LINUX AND WINDOWS OPERATING SYSTEMS

→ **AIM**: To study and execute basic networking commands in windows and linux operating systems for viewing network configuration, testing connectivity, checking routing information, monitoring network status and troubleshooting network-related issues.

→ **REQUIREMENTS**:
- Windows 10/11 or Linux (Ubuntu 20.04/22
- command prompt(windows)
- Linux
- Active network (LAN/ wifi)

- ip config / ifconfig / ip - displays network configuration
- ping - tests connectivity to a remote host.
- tracert / traceroute - Displays the path packets take t reach a destination.
- nslookup - Revolves domain name to IP addresses
- netstat - Displays active network connection ports.
- arp - Displays ARP cache
- hostname - Displays the system hostname.
- route - Displays or modifies the routing table.

→ **WINDOWS COMMANDS**:

- command: ipconfig

  Displays: IP address
  Subnet Mask
  Default gateway

- command: ipconfig /a

```
C:\Windows\system32>ping google.com

Pinging google.com [2404:6800:4007:83f::200e] with 32 bytes of data:
Reply from 2404:6800:4007:83f::200e: time=16ms
Reply from 2404:6800:4007:83f::200e: time=21ms
Reply from 2404:6800:4007:83f::200e: time=29ms
Reply from 2404:6800:4007:83f::200e: time=23ms

Ping statistics for 2404:6800:4007:83f::200e:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 16ms, Maximum = 29ms, Average = 22ms
```

```
C:\Windows\system32>nslookup google.com
Server:  UnKnown
Address:  192.168.23.81

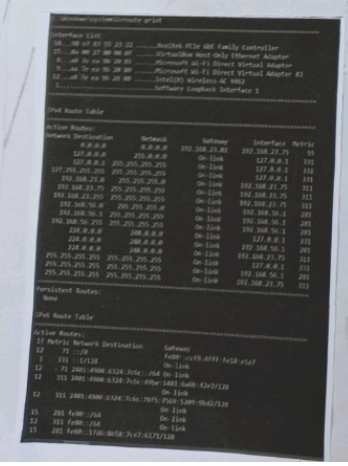
Non-authoritative answer:
Name:    google.com
Addresses:  2404:6800:4007:83f::200e
          172.217.24.206
```

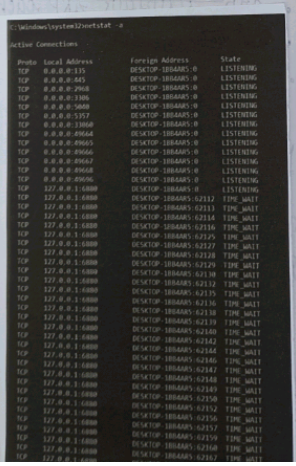
```
C:\Windows\system32>tracert google.com

Tracing route to google.com [2404:6800:4007:83f::200e]
over a maximum of 30 hops:

  1    3 ms     2 ms     5 ms  2401:4900:6324:7c6c::54
  2     *        *        *     Request timed out.
  3    28 ms    15 ms    16 ms  2401:4900:0:7f1::1
  4    45 ms    30 ms    25 ms  2401:4900:0:6fe::2
  5    20 ms    17 ms    22 ms  2401:4900:0:6f7::1
  6     *        *        *     Request timed out.
  7     *       36 ms    19 ms  2404:a800:1a00:1::601
  8    28 ms    12 ms     *     2404:a800::92
  9    16 ms    25 ms    21 ms  2001:4860:1:1::674
 10    48 ms    20 ms    20 ms  2001:4860:0:1::a039
 11    20 ms    21 ms    26 ms  2001:4860:0:1::33cb
 12    42 ms    28 ms    30 ms  prmaaa-at-in-x0e.1e100.net [2404:6800:4007:83f::200e]

Trace complete.
```





Lease information
Adapter Details

- Command: ping google.com
  Display: checks whether the destination is reachable.

- Command: tracert google.com
  purpose: Displays every router (hop) between your computer and destination.

- command: nslookup google.com
  Displays: DNS server
  IP address of website

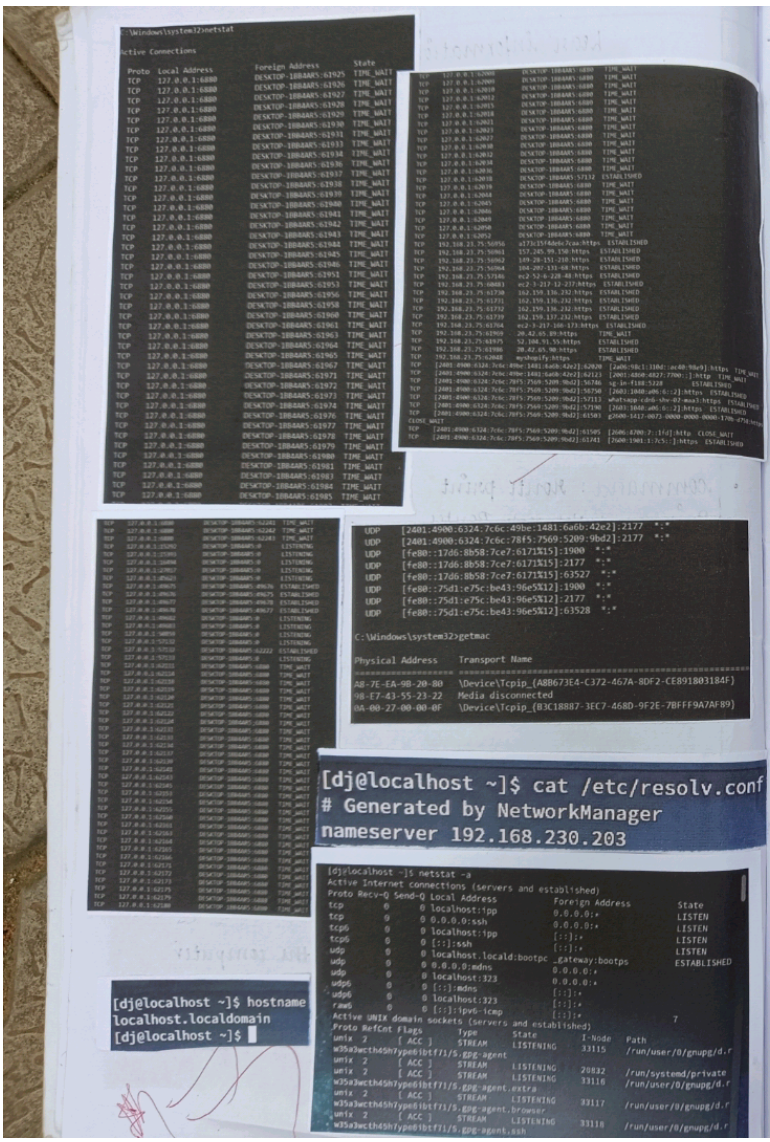
- command: route print
  Displays: Network Routes
  Gateway
  Destination Networks

- command: netstat
  displays: Active TCP connections
  Listening Ports

- command: netstat -a
  displays: All Ports
  TCP connections
  UDP ports

- command: getmac
  purpose: Displays MAC address of the computer

- command: host name
  purpose: displays computer name



```
C:\Windows\system32>netstat

Active Connections

  Proto  Local Address          Foreign Address        State
  TCP    127.0.0.1:6880         DESKTOP-1B8AAR5:61925  TIME_WAIT
  TCP    127.0.0.1:6880         DESKTOP-1B8AAR5:61926  TIME_WAIT
  TCP    127.0.0.1:6880         DESKTOP-1B8AAR5:61927  TIME_WAIT
  TCP    127.0.0.1:6880         DESKTOP-1B8AAR5:61928  TIME_WAIT
  TCP    127.0.0.1:6880         DESKTOP-1B8AAR5:61929  TIME_WAIT
  TCP    127.0.0.1:6880         DESKTOP-1B8AAR5:61930  TIME_WAIT
  TCP    127.0.0.1:6880         DESKTOP-1B8AAR5:61931  TIME_WAIT
  TCP    127.0.0.1:6880         DESKTOP-1B8AAR5:61933  TIME_WAIT
  TCP    127.0.0.1:6880         DESKTOP-1B8AAR5:61934  TIME_WAIT
  TCP    127.0.0.1:6880         DESKTOP-1B8AAR5:61936  TIME_WAIT
  TCP    127.0.0.1:6880         DESKTOP-1B8AAR5:61937  TIME_WAIT
  TCP    127.0.0.1:6880         DESKTOP-1B8AAR5:61938  TIME_WAIT
  TCP    127.0.0.1:6880         DESKTOP-1B8AAR5:61939  TIME_WAIT
  TCP    127.0.0.1:6880         DESKTOP-1B8AAR5:61940  TIME_WAIT
  TCP    127.0.0.1:6880         DESKTOP-1B8AAR5:61941  TIME_WAIT
  TCP    127.0.0.1:6880         DESKTOP-1B8AAR5:61942  TIME_WAIT
  TCP    127.0.0.1:6880         DESKTOP-1B8AAR5:61943  TIME_WAIT
  TCP    127.0.0.1:6880         DESKTOP-1B8AAR5:61944  TIME_WAIT
  TCP    127.0.0.1:6880         DESKTOP-1B8AAR5:61945  TIME_WAIT
  TCP    127.0.0.1:6880         DESKTOP-1B8AAR5:61946  TIME_WAIT
  TCP    127.0.0.1:6880         DESKTOP-1B8AAR5:61953  TIME_WAIT
  TCP    127.0.0.1:6880         DESKTOP-1B8AAR5:61956  TIME_WAIT
  TCP    127.0.0.1:6880         DESKTOP-1B8AAR5:61958  TIME_WAIT
  TCP    127.0.0.1:6880         DESKTOP-1B8AAR5:61960  TIME_WAIT
  TCP    127.0.0.1:6880         DESKTOP-1B8AAR5:61961  TIME_WAIT
  TCP    127.0.0.1:6880         DESKTOP-1B8AAR5:61963  TIME_WAIT
  TCP    127.0.0.1:6880         DESKTOP-1B8AAR5:61964  TIME_WAIT
  TCP    127.0.0.1:6880         DESKTOP-1B8AAR5:61965  TIME_WAIT
  TCP    127.0.0.1:6880         DESKTOP-1B8AAR5:61967  TIME_WAIT
  TCP    127.0.0.1:6880         DESKTOP-1B8AAR5:61971  TIME_WAIT
  TCP    127.0.0.1:6880         DESKTOP-1B8AAR5:61972  TIME_WAIT
  TCP    127.0.0.1:6880         DESKTOP-1B8AAR5:61973  TIME_WAIT
  TCP    127.0.0.1:6880         DESKTOP-1B8AAR5:61974  TIME_WAIT
  TCP    127.0.0.1:6880         DESKTOP-1B8AAR5:61976  TIME_WAIT
  TCP    127.0.0.1:6880         DESKTOP-1B8AAR5:61977  TIME_WAIT
  TCP    127.0.0.1:6880         DESKTOP-1B8AAR5:61978  TIME_WAIT
  TCP    127.0.0.1:6880         DESKTOP-1B8AAR5:61979  TIME_WAIT
  TCP    127.0.0.1:6880         DESKTOP-1B8AAR5:61980  TIME_WAIT
  TCP    127.0.0.1:6880         DESKTOP-1B8AAR5:61981  TIME_WAIT
  TCP    127.0.0.1:6880         DESKTOP-1B8AAR5:61983  TIME_WAIT
  TCP    127.0.0.1:6880         DESKTOP-1B8AAR5:61984  TIME_WAIT
  TCP    127.0.0.1:6880         DESKTOP-1B8AAR5:61985  TIME_WAIT
```

```
  UDP    [2401:4900:6324:7c6c:49be:1481:6a6b:42e2]:2177  *:*
  UDP    [2401:4900:6324:7c6c:78f5:7569:5209:9bd2]:2177  *:*
  UDP    [fe80::17d6:8b58:7ce7:6171%15]:1900  *:*
  UDP    [fe80::17d6:8b58:7ce7:6171%15]:2177  *:*
  UDP    [fe80::17d6:8b58:7ce7:6171%15]:63527  *:*
  UDP    [fe80::75d1:e75c:be43:96e5%12]:1900  *:*
  UDP    [fe80::75d1:e75c:be43:96e5%12]:2177  *:*
  UDP    [fe80::75d1:e75c:be43:96e5%12]:63528  *:*

C:\Windows\system32>getmac

Physical Address    Transport Name
=================== ==========================================================
A8-7E-EA-9B-20-B0   \Device\Tcpip_{A8667E4-C372-467A-8DF2-CE891803184F}
98-E7-43-55-23-22   Media disconnected
0A-00-27-00-00-0F   \Device\Tcpip_{B3C18887-3EC7-468D-9F2E-7BFFF9A7AF89}
```

```
[dj@localhost ~]$ cat /etc/resolv.conf
# Generated by NetworkManager
nameserver 192.168.230.203
```

```
[dj@localhost ~]$ netstat -a
Active Internet connections (servers and established)
Proto Recv-Q Send-Q Local Address           Foreign Address         State
tcp        0      0 localhost:ipp           0.0.0.0:*               LISTEN
tcp        0      0 0.0.0.0:ssh             0.0.0.0:*               LISTEN
tcp6       0      0 localhost:ipp           [::]:*                  LISTEN
tcp6       0      0 [::]:ssh                [::]:*                  LISTEN
udp        0      0 localhost.localdo:bootpc _gateway:bootps        ESTABLISHED
udp        0      0 0.0.0.0:mdns            0.0.0.0:*
udp6       0      0 localhost:323           [::]:*
udp6       0      0 [::]:mdns               [::]:*
raw6       0      0 [::]:ipv6-icmp          [::]:*                  7
Active UNIX domain sockets (servers and established)
Proto RefCnt Flags       Type       State         I-Node   Path
unix  2      [ ACC ]     STREAM     LISTENING     33115    /run/user/0/gnupg/d.r
w35a3wcth45h7ype61btf71/S.gpg-agent
unix  2      [ ACC ]     STREAM     LISTENING     20832    /run/systemd/private
unix  2      [ ACC ]     STREAM     LISTENING     33116    /run/user/0/gnupg/d.r
w35a3wcth45h7ype61btf71/S.gpg-agent.extra
unix  2      [ ACC ]     STREAM     LISTENING     33117    /run/user/0/gnupg/d.r
w35a3wcth45h7ype61btf71/S.gpg-agent.browser
unix  2      [ ACC ]     STREAM     LISTENING     33118    /run/user/0/gnupg/d.r
w35a3wcth45h7ype61btf71/S.gpg-agent.ssh
```

```
[dj@localhost ~]$ hostname
localhost.localdomain
[dj@localhost ~]$
```

- command: arp -a
- purpose: Displays IP-to-MAC address mappings

- command: netstat -a
- purpose: All ports
  TCP connections
  UDP ports

→ LINUX NETWORKING COMMANDS:

- command: ip addr or ifconfig
- purpose: Displays IP address

- command: ip route
- purpose: Shows • default Gateway
  • Routing Table

- command: hostname
- purpose: displays system hostname

- command: ping google.com

- command: traceroute google.com
  (If not installed:)
  sudo apt install traceroute

- command: netstart -a (or) ss -a
- purpose: display network statistics

- command: arp -a (or) ip neigh
- purpose: Display ARP table

- A command: ip link
- purpose: display MAC address

```
[dj@localhost ~]$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group defaul
t qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
       valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP gr
oup default qlen 1000
    link/ether 08:00:27:99:a3:3d brd ff:ff:ff:ff:ff:ff
    inet 10.0.2.15/24 brd 10.0.2.255 scope global dynamic noprefixroute enp0s3
       valid_lft 86313sec preferred_lft 86313sec
    inet6 fe80::a00:27ff:fe99:a33d/64 scope link noprefixroute
       valid_lft forever preferred_lft forever
[dj@localhost ~]$
```

```
[dj@localhost ~]$ ip route
default via 10.0.2.2 dev enp0s3 proto dhcp src 10.0.2.15 metric 100
10.0.2.0/24 dev enp0s3 proto kernel scope link src 10.0.2.15 metric 100
[dj@localhost ~]$
```

```
[dj@localhost ~]$ nslookup google.com
Server:         192.168.230.203
Address:        192.168.230.203#53

Non-authoritative answer:
Name:   google.com
Address: 142.251.222.174
Name:   google.com
Address: 2404:6800:4007:83b::200e
```

```
[dj@localhost ~]$ ip link
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN mode DEFAULT
 group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP mo
de DEFAULT group default qlen 1000
    link/ether 08:00:27:99:a3:3d brd ff:ff:ff:ff:ff:ff
[dj@localhost ~]$
```

```
[dj@localhost ~]$ ss -tuln
Netid   State    Recv-Q   Send-Q     Local Address:Port      Peer Address:Port
udp     UNCONN   0        0              0.0.0.0:5353           0.0.0.0:*
udp     UNCONN   0        0            127.0.0.1:323            0.0.0.0:*
udp     UNCONN   0        0                [::]:5353              [::]:*
udp     UNCONN   0        0               [::1]:323               [::]:*
tcp     LISTEN   0        4096         127.0.0.1:631            0.0.0.0:*
tcp     LISTEN   0        128            0.0.0.0:22             0.0.0.0:*
tcp     LISTEN   0        4096             [::1]:631              [::]:*
tcp     LISTEN   0        128              [::]:22                [::]:*
[dj@localhost ~]$
```

```
[dj@localhost ~]$ curl google.com
<HTML><HEAD><meta http-equiv="content-type" content="text/html;charset=utf-8">
<TITLE>301 Moved</TITLE></HEAD><BODY>
<H1>301 Moved</H1>
The document has moved
<A HREF="http://www.google.com/">here</A>.
</BODY></HTML>
[dj@localhost ~]$
```

- command : cat /etc/resolv.conf
- displays : DNS configuration

- command: ip link show
- displays : • UP/DOWN status
  • Interface name
- purpose: Check interface status

- command: ss -tuln
- purpose : Displays : • TCP ports
  • UDP ports
  • listening services

- command: curl google.com (or) wget google.com
- purpose: To check internet connectivity.

- command: curl google.com (or) wget google.com
- purpose : check internet connectivity

→ RESULT: The basic networking commands in linux & windows operating systems has been implemented and executed successfully.